

JAVKHLANTUGS TSEDENSUMIYA

MACHINE LEARNING ENGINEER

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SUMMARY

Machine Learning Engineer with 3+ years building end-to-end AI systems (NLP, Computer Vision, RAG, multimodal). Expert in Python, PyTorch & TensorFlow pipelines from scraping to deployment. Former mobile startup founder and surgeon bringing production engineering and analytical rigor to ML solutions.

TECHNICAL SKILLS

Python, PyTorch, TensorFlow, Hugging Face, LangChain, Ollama • LLMs, RAG, SwinIR, OCR, Speech Recognition, AI Agents • Pandas, NumPy, OpenCV, ChromaDB, SQL, MongoDB, Flask, Streamlit, Playwright • Docker, Git, Model Optimization

EXPERIENCE

AI and Machine Learning Engineer (Independent | Freelance) 2023 - Present

- Built full ML pipelines: web scraping, preprocessing, training, evaluation & deployment (PyTorch/TensorFlow).
- Developed RAG systems and LLM applications for domain-specific QA and knowledge retrieval.
- Created Computer Vision models (SwinIR restoration, OCR, object detection, medical X-ray) with data augmentation.
- Engineered scalable pipelines (SQL, MongoDB, ChromaDB) optimized for low-resource hardware.
- Optimized models via tuning, cross-validation, error analysis and metrics.

Data Analyst (Tes Petrolium) 2022 - 2023

- Applied ML models and statistical analysis on business data for predictive insights.
- Automated preprocessing, feature engineering, visualization and reporting pipelines (Python/SQL).

Founder & Mobile Developer (CEO) Joot LLC 2013 - 2022

- Founded and scaled mobile app startup delivering production Android/iOS apps (Swift, Kotlin, Java, Flutter).
- Managed full lifecycle: architecture, development, deployment, maintenance and monetization.

General Surgery Resident (Mongolian National University of Medical Sciences) 2009 - 2012

- Completed clinical residency focused on diagnostic reasoning and high-pressure decision-making.
- Developed analytical skills transferable to ML experimentation and model debugging

PROJECTS

VecRAG: Local RAG system with ChromaDB, LangChain, and Ollama for semantic search.

Image Enhancer: Flask web app for Swin Transformer image restoration (PyTorch).

Mock Interviewer: Fully local AI agent with TTS/STT and real-time LLM feedback.

PneumoScan: EfficientNet model for pneumonia detection from chest X-rays (Streamlit).

EDUCATION

Bachelor of Medicine & General Surgeon

- Mongolian National University of Medical Sciences, Graduated: July 2009

CERTIFICATIONS

TensorFlow Developer Certificate • IELTS 6.0